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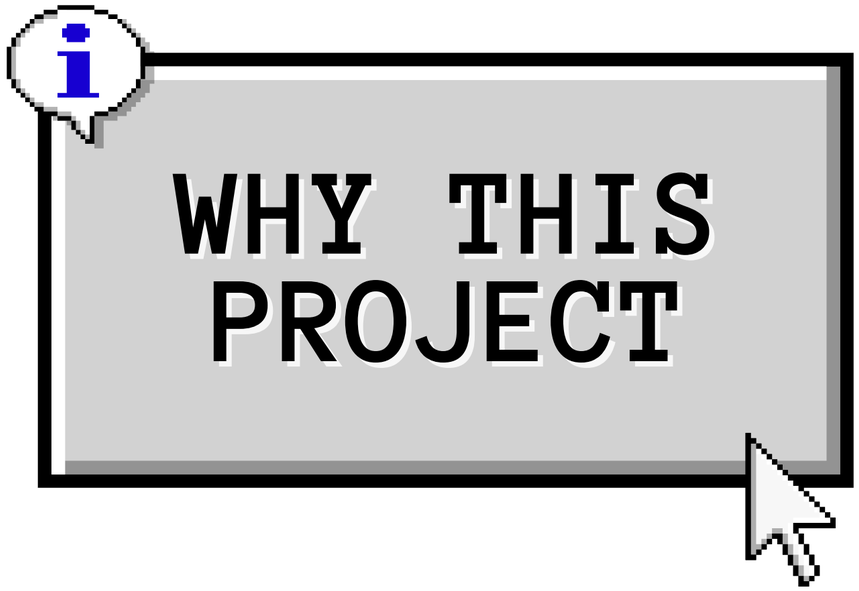
AMD Developer Hackathon · ACT III

# WORKSIGHT

DEFECTS FROM GOOD PHOTOS



Start

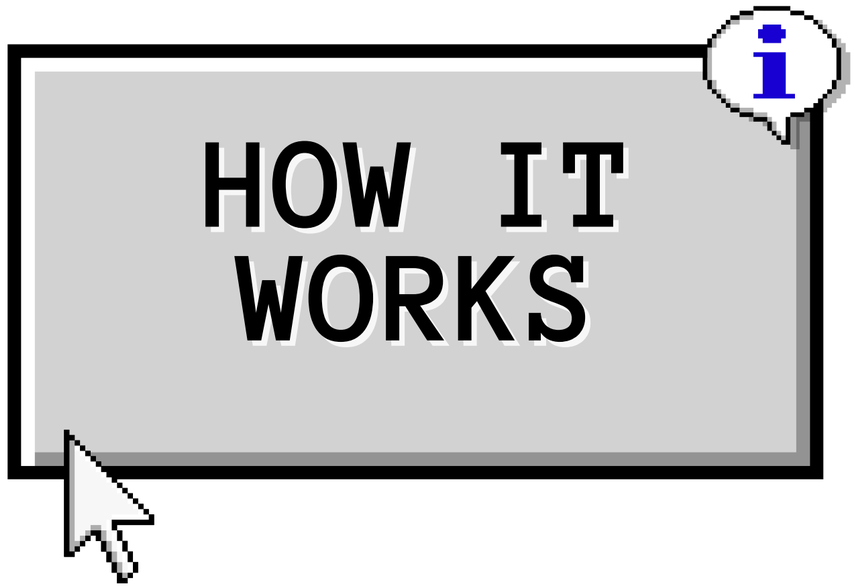



A new part on the line has no defect dataset. Defects are rare, so collecting and labelling hundreds of them takes weeks, and every new part or camera view starts again. Worksight starts from what every line already has: photos of good parts.

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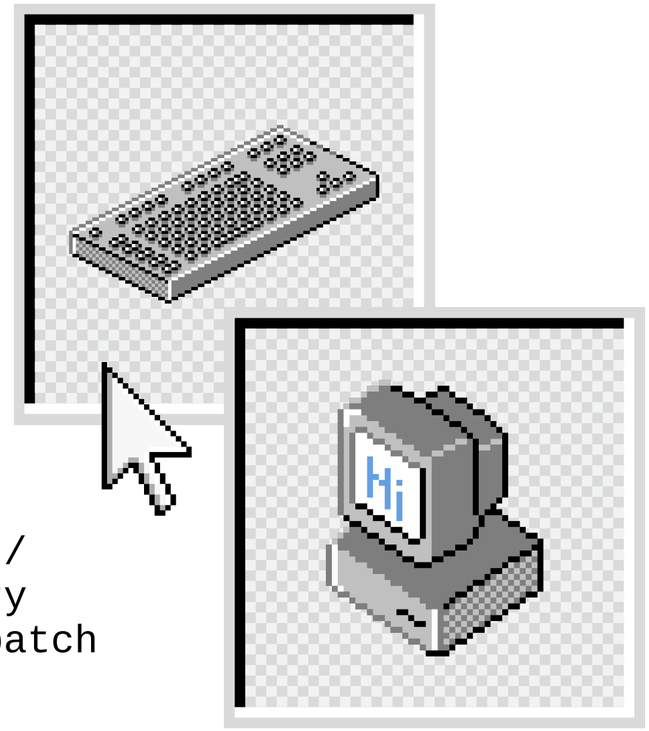
1. 20 photos of good parts build a profile in about 9 s. No labels, no training.
2. Each part gets an anomaly score, a heatmap and the closest good reference.
3. Gemma on AMD explains what to check. The engineer decides.



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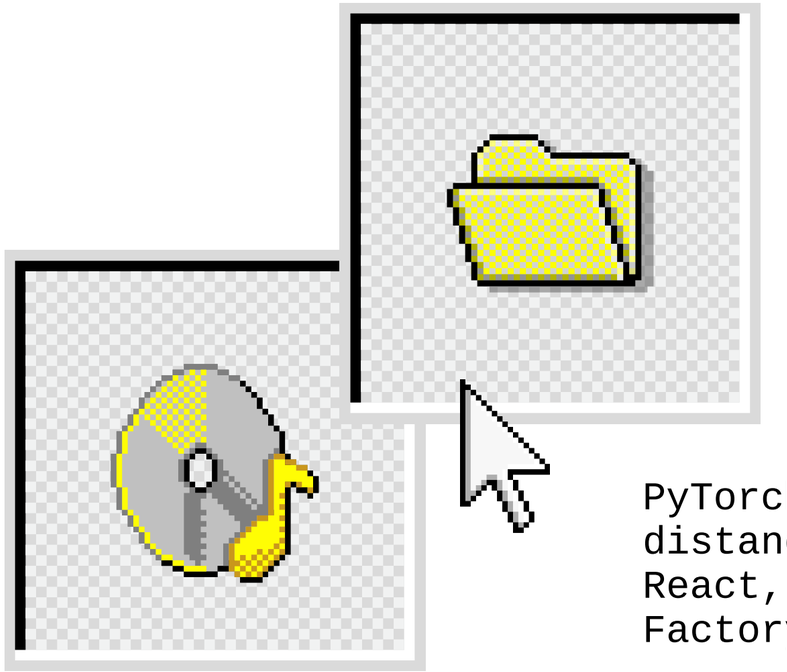
# THE LIVE PRODUCT




Algorithm, AI and engineer on one screen: heatmap and closest reference, Pass / Fail / Needs review, Gemma's explanation with every inspection, one-page PDF report, history, batch inspection, 7 languages.

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# TECH STACK

PyTorch: frozen ResNet-18 features, nearest-patch distance. FastAPI, SQLite, job queue. Next.js 16, React, TypeScript. Gemma-4-31B-IT on AMD Token Factory. Railway. pytest, Vitest, Playwright.

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# BENCHMARKS

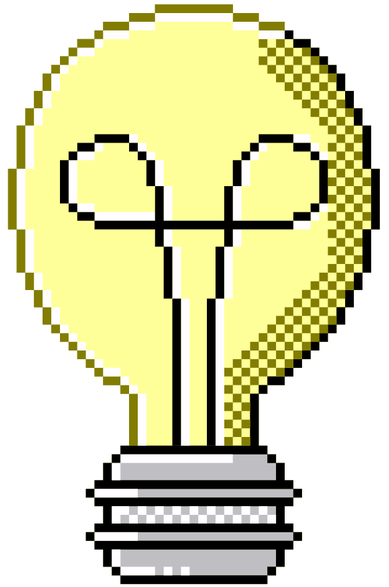
VisA PCB · held-out test · 20 good references · CPU

| Set | AUROC | F1 | Recall | False alarms | Defects to engineer | Good to engineer | p50 |
|---|---|---|---|---|---|---|---|
| **pcb1** | 0.905 | 0.591 | 44% | 5% | **56%** | 7% | 250 ms |
| **pcb2** | 0.884 | 0.715 | 59% | 6% | **77%** | 13% | 304 ms |
| **pcb3** | 0.875 | 0.613 | 46% | 4% | **58%** | 10% | 250 ms |
| **pcb4** | 0.992 | 0.921 | 87% | 2% | **96%** | 4% | 194 ms |

Recall = defects above the threshold. "To engineer" counts the review band (within 10% of the threshold). Mean AUROC 0.914. CPU run, not an AMD GPU. Source: docs/BENCHMARK.md

# PARTNER TECH & WHY

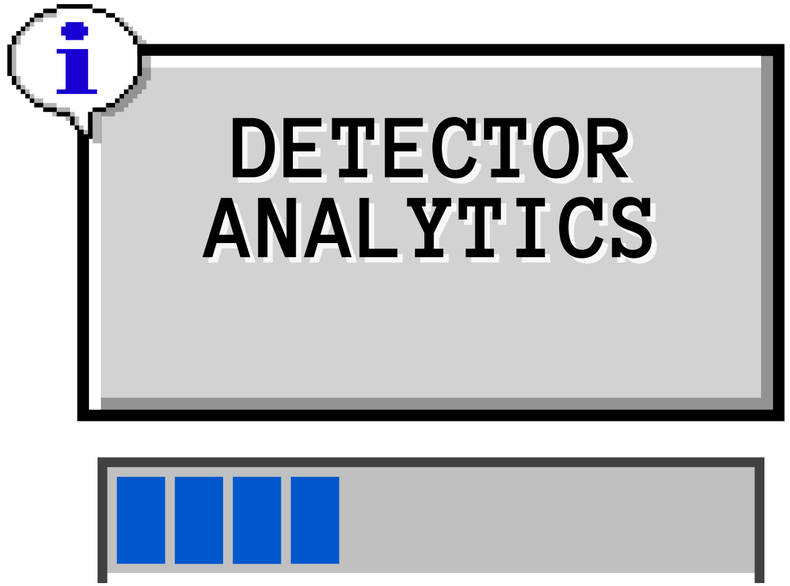
AMD Token Factory runs Gemma on AMD GPUs; the API key stays on our server. Gemma 4 31B reads the photo, heatmap, reference and part documents and explains in plain words. DeepSeek V4.1 Flash is the fallback. Next: the detector on AMD Instinct.



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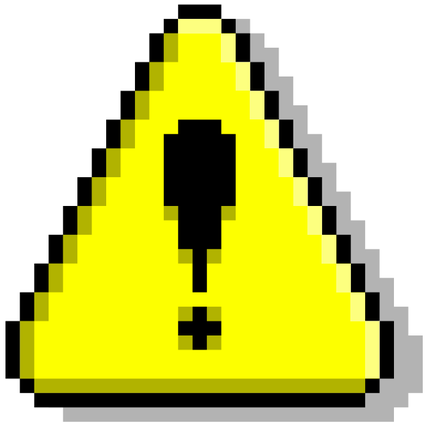
AUROC 0.88 to 0.99 on four circuit-board sets. F1 0.59 to 0.92.




v2 raised dev AUROC from 0.806 to 0.862. A 24-variant sweep found nothing better, so production stayed unchanged.

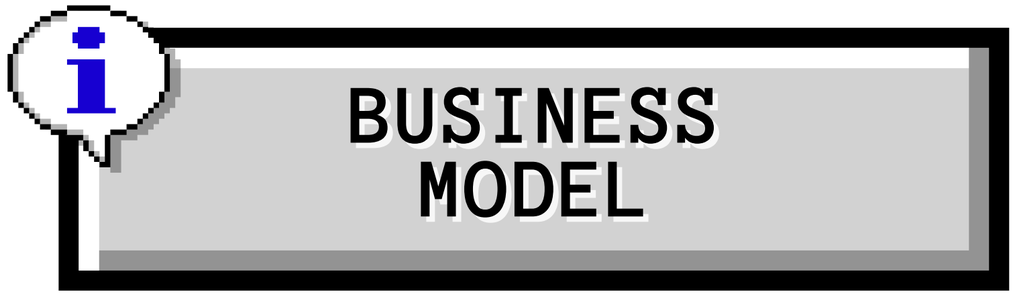
The engineer decides; model and engineer decisions are stored separately. Gemma never changes the detector result. A new profile version goes live only if it adds no false alarm or missed defect, and it can be rolled back.








AI visual inspection is forecast to grow by about $27B by 2029 (Technavio). Most tools need labelled defects or are code libraries. Worksight starts from good photos, explains each result and keeps the engineer in charge.



# BUSINESS MODEL

Yes

Cancel

Subscription per camera view, or on-premises on AMD hardware for plants that keep images on site. Pilot on one line and one camera: measure engineer minutes per part and escaped defects.

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Bakhtiyar, CEO and AI engineer: detector, benchmarks, Gemma integration. Ali, frontend developer: operator workspace and reports. Next: detector on AMD Instinct, team workspaces, edge station on AMD Ryzen AI.



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# TRY IT NOW

Scan to open the live demo with the AI description, or connect with us on LinkedIn.

